

- **Academic** with keen interest in communicating, demystifying, and contextualizing advanced concepts to promote a holistic understanding of technology.
- **Researcher** working in the computer networks and information security domain. Specific focus on the confluence of emerging technologies including Internet-of-Things, cryptocurrencies, blockchain, machine learning, and virtual environments. Emphasis on high-quality novel and interdisciplinary research that interrogates new paradigms.
- **Research consultant** with track record of advising government and citizen organizations on technology policy and transformation. Strong advocate of responsible and sustainable use of technology.
- **Election technology practitioner**

EXPERIENCE

Associate Professor, Electrical Engineering, National University of Sciences and Technology 2023-

Assistant Professor, Electrical Engineering, National University of Sciences and Technology 2016-2023

- I teach in the NUST School of Electrical Engineering and Computer Science (SEECs)
- Courses include computer programming, computer networks, network security, and cryptocurrencies.
- I head the Cognet Lab, where we research security applications in new and emerging technologies, including blockchain, machine learning, computer vision, Internet of Things, etc.

Research Associate, Computing Science, Newcastle University, UK 2014-2016

- Flagship EU project on secure and verifiable electronic voting systems
- Contributed to projects on cryptocurrencies and blockchain
- Initiated a project on [next-generation botnets](#)
- Contributed to a book: [Real-World Electronic Voting: Design, Analysis and Deployment](#)

Postdoctoral Researcher, University of New South Wales, Australia 2012-2014

- Project on security and green applications of software-defined networks
- Data provenance mechanisms for healthcare and IoT devices

ICT Research Projects Officer, CSIRO, Australia 2011-2012

- I worked on a project to 'green' residential WiFi networks. We developed algorithms to strategically adjust the mode and transmit power of WiFi access points in response to user traffic, with power savings of up to 40%. We showcased a working demo in June, 2012.

Assistant Manager, WorldCALL Broadband, Pvt. Ltd (WBL), Pakistan 2003-2005

- I took on various engineering and management roles in the Engineering department in deploying Pakistan's largest hybrid-fiber-coax broadband network.

EDUCATION

● **PhD in Electrical Engineering, University of New South Wales (UNSW), Australia** 2007-2012

- **Thesis:** [Security Mechanisms for Body Sensor Networks](#)
- **Supervisors:** Vijay Sivaraman (UNSW), Diethelm Ostry (CSIRO)
- My dissertation identifies and proposes novel solutions for key security requirements in the emerging paradigm of bodyworn health monitoring devices, including secure broadcast, data confidentiality, data authentication and provenance. Traditional security solutions in these domains are too resource and compute-intensive for bodyworn devices, necessitating the need for new efficient and lightweight solutions, including use of unorthodox non-crypto primitives.
- My work is validated with extensive experimentation in varied environments, including indoor living and work spaces, experimental anechoic chambers, and outdoor settings. In one case, we even fitted sensor nodes on members of a local first division football league.
- A short video interview may be viewed at <http://youtu.be/DbTy8exjEog>

MS in Telecommunications Engineering, University of New South Wales (UNSW), Australia 2005-2006

- Courses included Network Management, Advanced Computer Networks (sensor networks, game theory, etc.), Switching Systems and Design, Satellite Systems and Navigation, and Wireless Communications.

BS in Electrical Engineering, GIK Institute of Eng. Sciences & Tech. Pakistan 1998-2002

- Final Year Project: [Software Control for Embedded Systems using Jini](#)

CONSULTING & PROJECTS

● **Education: 2 year Coding Fellows Program** 2024-

- I am part of a small team managing a project to train 75 teachers (Coding Fellows) to deliver STEM education (Coding, AI, IT, and Information Security) in 150 government schools within Islamabad. The project is funded by the Ministry of Education. I am developing and curating lesson plans, organizing STEM competitions and tournaments between different schools, and introductory university visits for students.

Education: 6 month Bootcamp on Blockchain Technology – Gilgit Baltistan 2023

- I am currently curating and teaching a 6-month specialized course in blockchain technology aimed at enhancing skillset of youth in Gilgit Baltistan and supporting the local IT and freelancing community.
- This project, titled, **High Impact Skills Development Program in AI, Data Science, and Blockchain**, is an ongoing flagship collaboration between the Gilgit Baltistan government and NUST University.

Research Consultant: Election Technologies and Best Practices – UNDP 2022-2023

- I authored concept notes and developed roadmaps for introducing electronic voting machines in Pakistan
- I assisted the Election Commission of Pakistan (ECP) with plans to set up an in-house R&D Center.
- I advised the ECP on various matters including modernization of processes, overseas voting, stakeholder outreach, staff training programs, and election monitoring strategies, etc.
- I liaised with research institutes, universities, and civil society groups to collaborate with ECP on projects.

Research: Automating Industrial Supply Chains using a Blockchain-based IoT Solution – HEC 2018-2023

Research: *Electronic Voting Machines for Pakistan* – RASTA

2021-2022

- I served as Co-PI on a project to devise a policy framework to introduce electronic voting in Pakistan and devise a comprehensive roadmap for stakeholders aligned with international best practices.
- This project was funded by the Pakistan Institute of Development Economics as part of its competitive RASTA grants program to produce high quality, evidence-based policy research in Pakistan.
- Project report: [*Electronic Voting Machines for Pakistan: Opportunities, Challenges, and the Way Forward*](#)

Consultant: *Technology for Efficient and Effective Elections (T3E)* – TDEA

2019

- I worked on a project to research and improve use of technology in Pakistan's electoral processes.
- This project was headed by Trust for Democratic Education and Accountability (TDEA), a prominent non-profit organization working in the elections observation domain.
- I served as member of the Election Tech Forum, a gathering of experts and civil society personnel convened to support this project with technical expertise.

TEACHING & ACADEMICS

- I teach the following courses routinely at NUST-SEECS
 - **Undergrad Courses:** Fundamentals of ICT, Fundamentals of Programming, Computer Networks.
 - **Masters Courses:** Computer Networks, Network Routing and Switching, Advanced Computer Networks, Selected Topics in Networks, Cryptocurrencies and Blockchain.
 - I usually receive student feedback in the >85% range.
- **Research students:** 25 Masters students and 2 PhD students have graduated under my supervision.
- I introduced a **Masters-level course on Cryptocurrencies and Blockchain technology** at NUST in 2018, the first in the country and among the earliest such offerings in the world. This course is popular and continues to run across multiple departments.
- I am curating a 6-month high intensity blockchain technology boot-camp in Gilgit Baltistan. Course contents include: cryptocurrencies, smart contracts, Solidity essentials, decentralized applications, security, policy, and various advanced topics.
- I have designed and conducted mini-courses (5-10 hours) which include: History of the Internet, Electronic Voting – Technology to Policy, and Demystifying Cryptocurrencies and Blockchain.
- I can teach courses in the security domain, including Cryptography 101, Security Engineering, Information Security Management, Network Security, etc.

OUTREACH & ADVOCACY

Policy Consultations & Committees

2021-2022

- I served as Member of the President's Subcommittee on Emerging Technologies.
- I served on various committees for the Ministry of IT & Telecommunications and the Ministry of Science and Technology.
- I advised a Parliamentary consultation on Digital rights, Assets and Cryptocurrency.
- I served on the Election Commission's Technical Committee responsible for devising policy on electronic voting and assessing different electronic voting machines.

OneVote Annual Conference

2021

- I co-curated the [One Vote Annual Conference on Rights, Access, and Technology in Elections](#)
- This regional conference was organized by Indian civil society organization, Hasgeek, and was a two day event featuring technology experts, policy makers, and scholars.
- I delivered a talk titled [How a State Examines Technology](#) on Pakistan's election technology experience.

PIVOT (*Promoting Innovative Voting Technologies*) Project

2021



- Our NUST research group launched PIVOT to promote responsible use of election technology.
- This initiative involved technical talks, newspaper articles, a Twitter [presence](#), and a dedicated [newsletter](#)
- We undertook the first ever study in Pakistan on the dynamics of overseas voting: [The Game-Changing Overseas Vote](#)

Opinion Pieces, Cover Stories, Reviews, and Talks

2015-

- I frequently contribute articles to local media outlets, most frequently Pakistan's leading national English papers, [DAWN](#) and [Express Tribune](#). These include op-eds, cover stories on science, book reviews, etc.
- I frequently give guest talks and presentations on technology-related topics to government organizations, civil society organizations, civil society groups, and on local TV shows.
- Our research on next-generation Bitcoin-powered botnets featured in [Forbes magazine](#)

Google Scholar <https://scholar.google.com/citations?user=0du4338AAAAJ&hl>



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PUBLICATIONS

BOOK CHAPTERS

- S. Taha Ali, J. Murray, **An Overview of End-to-end Verifiable Voting Systems**, Real-World Electronic Voting: Design, Analysis and Deployment, 2016 (CRC Press)

CONFERENCES

- H. B. Haq, S. Taha Ali, SF. Shahandashti, **Cuckoo's Nest: An Ultra-Lightweight DoS-resilient Bitcoin Mempool**, 7th IEEE International Conference on Blockchain and Cryptocurrency, ICBC, 2025
- H. B. Haq, S. Taha Ali, R. McDermott, **End-to-end Verifiable Voting for Developing Countries - What's Hard in Lausanne is Harder Still in Lahore**, International Conference for Electronic Voting, E-Vote-ID, 2022
- A. Wajid, N. Kamal, M. Sharjeel, R. M. Sheikh, H. B. Wasim, M. H. Ali, W. Hussain, S. Taha Ali, L. Anjum. **A First Look at Private Communications in Video Games using Visual Features**, Privacy Enhancing Technologies, 2021
- H. B. Haq, R. McDermott, S. Taha Ali, **Pakistan's Internet Voting Experiment**, International Conference for Electronic Voting E-Vote-ID, 2019 [**Best Paper Award, Practitioners Track**]
- P. McCorry, M. Möser, S. Taha Ali, **Why Preventing a Cryptocurrency Exchange Heist Isn't Good Enough** Cambridge Security Protocols Workshop, 2018
- D. Clarke, S. Taha Ali, **End to End Security is Not Enough**, Cambridge Security Protocols Workshop, 2017
- M. Siddiqi, S. Taha Ali, V. Sivaraman, **Secure Lightweight Context-driven Data Logging for Bodyworn Sensing Devices**, International Symposium on Digital Forensics and Security (ISDFS), 2017
- S. Taha Ali, D. Clarke, P. McCorry, **Bitcoin: Perils of an Unregulated Global P2P Currency**, Cambridge Security Protocols Workshop, 2015
- S. Taha Ali, P. McCorry, Peter HJ Lee, F. Hao, **ZombieCoin: Powering Next Generation Botnets with Bitcoin**, BITCOIN, Financial Cryptography (FC), 2015
- S. Taha Ali, V. Sivaraman, D. Ostry, S. Jha, **Securing Data Provenance in Body Area Networks using Lightweight Wireless Link Fingerprints**, TrustED Workshop, ACM Computer and Communications Security (CCS), 2013
- S. Taha Ali, V. Sivaraman, D. Ostry, **Zero Reconciliation Secret Key Generation for Mobile Bodyworn Health Monitoring Devices**, ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec), 2012.
- S. Taha Ali, V. Sivaraman, D. Ostry, **Authentication of Lossy Data in Body Sensor Networks**, IEEE International Conference on Sensing, Communication, and Networking (SECON), 2012
- L. Yao, S. Taha Ali, V. Sivaraman, D. Ostry, **Decorrelating Secret Bit Extraction via Channel Hopping in Body Area Networks**, IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), 2012
- L. Yao, S. Taha Ali, V. Sivaraman, D. Ostry, **Improving secret key generation performance for on-body devices**, EAI Body Area Networks (BodyNets), 2011
- S. Taha Ali, V. Sivaraman, D. Ostry, **Secret Key Generation Rate vs. Reconciliation Cost Using Wireless Channel Characteristics in Body Area Networks**, IEEE/IFIP International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), 2010.
- S. Taha Ali, V. Sivaraman, A. Dhamdhere, **A Per-Hop Security Scheme For Highly Dynamic Wireless Sensor Networks**, IEEE Mobile Adhoc and Sensor Systems (MASS), 2009
- S. Taha Ali, V. Sivaraman, A. Dhamdhere, D. Ostry, **Secure Key Loss Recovery For Network Broadcast In Single-Hop Wireless Sensor Networks**, IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), 2008

JOURNALS

- H. B. Haq, S. Taha Ali, SF. Shahandashti, **Neonpool: Reimagining Cryptocurrency Transaction Pools for Lightweight Clients and IoT Devices**, (under submission) [arxiv preprint] 2025
- A. Ahmad, A. Ahmed, A. Al-Dweik, S. Taha Ali, and A. Ahmad, **Merging Engine Implementation for Intra-frame Sharing in Multi-tenant Virtual Passive Optical Networks**, Journal of Optical Communications and Networking (JOCN), 2023
- R. Safdar, A. Pasha, S. Taha Ali, **False Data Injection Attacks on Networked Control Systems**, Journal of Control and Decision, 2023
- A. Tariq, H. B. Haq, S. Taha Ali, **Cerberus: A Blockchain-Based Accreditation and Degree Verification System**, IEEE Transactions on Computational Social Systems (TCSS), 2023
- M. Siddiqi, S. Taha Ali, V. Sivaraman, **Secure Opportunistic Contextual Logging for Wearable Healthcare Sensing**, IEEE Transactions on Dependable and Secure Computing, 2021
- M. Siddiqi, S. Taha Ali, V. Sivaraman, **Forensic Verification of Health Data From Wearable Devices Using Anonymous Witnesses**, IEEE Internet Things Journal, 2020
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- S. Taha Ali, V. Sivaraman, A. Radford, S. Jha, **A Survey of Securing Networks using Software Defined Networking**, IEEE Transactions on Reliability, 2015
- S. Taha Ali, V. Sivaraman, D. Ostry, S. Jha, G. Tsudik, **Securing First-Hop Data Provenance for Bodyworn Devices using Wireless Link Fingerprints**, IEEE Transactions on Information Forensics and Security (TIFS), 2015
- V. Sivaraman, J. Matthews, C. Russell, S. Taha Ali, A. Vishwanath, **Greening Residential WiFi Networks under Centralized Control**, IEEE Transactions on Mobile Computing (ToMC), 2015
- S. Taha Ali, V. Sivaraman, D. Ostry, **Authentication of Lossy Data in Body-sensor Networks for Cloud-based Healthcare Monitoring**, Future Generation Computer Systems, 2014
- S. Taha Ali, V. Sivaraman, D. Ostry, **Eliminating Reconciliation Cost in Secret Key Generation for Bodyworn Health Monitoring Devices**, IEEE Transactions on Mobile Computing, 2014
- S. Taha Ali, V. Sivaraman, A. Dhamdhere, D. Ostry, **Secure Key Loss Recovery for Network Broadcast in Single-hop Wireless Sensor Networks**, Elsevier Ad hoc Networks, 2010

OTHERS

- H. B. Haq, S. Taha Ali, **Electronic Voting Machines for Pakistan: Opportunities, Challenges, and the Way Forward**, PIDE Research for Social Transformation and Advancement (RASTA) Conference, 2022 [Policy Paper]
- Internet Voting Task Force, **Findings And Assessment Report of Internet Voting Task Force On Voting Rights Of Overseas Pakistanis**, 2018 [assessment presented to the Supreme Court of Pakistan]
- H. B. Haq, S. Taha Ali, **Navigating the Cryptocurrency Landscape: An Islamic Perspective**, 6th International Conference on Islam and Liberty (ILN), 2018 [conference/book chapter]
- S. Taha Ali, D. Clarke, P. McCorry, **The Nuts and Bolts of Micropayments: a Survey**, 2017 [arXiv preprint]
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- S. Taha Ali, **Security Mechanisms for Body Sensor Networks**, University of New South Wales, Australia, 2011 [PhD Thesis]

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